TIC-TAC-TOE AI USING PYTHON

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- **INTERNSHIP DOMAIN:**
  Artificial Intelligence

- **ORGSNIZATION:**

  CODSOFT

 # PROJECT: TIC-TAC-TOE Using python

INTRODUCTION

This project is a Python-based Tic-Tac-Toe game developed as part of the CODSOFT Artificial Intelligence Internship Program. The project allows users to paly Tic-Tac-Toe in a command-line interface environment.

The game demonstrates basic Artificial Intelligence concepts, game logic implementation, and user interaction handling using Python programming.

OBJECTIVE

The main objective of this project is:

- To develop an AI based Tic-Tac-Toe game using python.
- To understand game logic and decision-making concepts.
- To improve problem-solving and programming skills.

TECHNOLOGY USED

- Python – Used for implementing the game logic and functionality.

- Vs Code – Used as the code editor for writing and executing the program.

FEATURES

- Human vs Human gameplay
- Interactive command-line interface
- Winner detection system
- Draw detection system
- Easy-to-Use gameplay

SYSTEM REQUIRED

- **Hardware requirements:**
  -RAM: Minimum GB
  -Storage: 500MB free space
  -Computer/Laptop
  -Processor: Intel i3 or above

- **Software:**
  -Python 3.x
  -VS code or any Python IDE
  -Command Prompt/ Terminal

ALGORITHM

- **Step 1**
  Star the game.
- **Step 2**
  Create a 3*3 Tic-Tac-Toe board.
- **Step3**
  Assign Player X and Player 0.
- **Step 4**
  Display the game board on the screen.
- **Step 5**
  Ask the current player to enter the position (1-9).

- **Step 6**
  Check weather the entered position is valid or not.
- **Step 7**
  Update the board with the player's symbol.
- **Step 8**
  Check for winning conditions:
  - Horizontal match
  - Vertical match
  - diagonal match
- **Step 9**
  If a player wins, display the winner and end the game.
- **Step 10**
  If all positions are filled, and no player wins, declare the match as a Draw.
- **Step 11**
  Otherwise switch the turn to the next player.
- **Step 12**
  Repeat the process until game ends.

SOURCE CODE

```
board = [" " for _ in range(9)]

def print_board():
    print()
    for i in range(3):
        print(board[i*3] + " | " + board[i*3+1] + " | " +
board[i*3+2])
```

```
        if i < 2:
            print("--+---+--")
    print()

def check_winner(player):
    win_combination = [
        (0,1,2),(3,4,5),(6,7,8),
        (0,3,6),(1,4,7),(2,5,8),
        (0,4,8),(2,4,6)
    ]
    for combo in win_combination:
        if board[combo[0]] == board[combo[1]] ==
board[combo[2]] == player:
            return True
    return False

def is_full():
    return " " not in board

def play_game():
    current_player = "X"

    while True:
        print_board()
        move = int(input(f"player {current_player}, enter
position (1-9): ")) -1

        if board[move] != " ":
            print("Position already taken! Try again.")
```

```
        continue

    board[move] = current_player

    if check_winner(current_player):
        print_board()
        print(f"player {current_player} wins!")
        break

    if is_full():
        print_board()
        print("It's a draw!")
        break

    current_player = "0" if current_player == "X" else
"X"

play_game()
```

OUTPUT

The program successfully runs in the Terminal and allows two players to play the Tic-Tac-Toe game interactively.

Sample Output

```
X | 0 |
--+---+--
X | 0 | 0
--+---+--
```

X | | X   player X wins!

The game displays:

- Updated board after every move.
- Winner message when a player wins.
- Draw message if no winner is found.

LEARNING OUTCOMES

Through this project I have learned:

- Basics of Artificial Intelligence
- Python programming concepts
- Game logic implementation
- User input handling
- Problem-solving techniques

FUTURE ENHANCEMENT

- Add Graphical user Interface (GUI)
- Implement Minimax Algorithm
- Add Multiplayer functionality
- Convert into a web-based game
- Add difficulty levels

CONCLUSION

The Tic-Tac-Toe AI project helped in understanding basic AI concepts and game development using Python. It improves logical thinking and programming skills while creating and interactive gaming application.

REFERENCES

- Python Official Documentation
- Online Tutorials and Learning resources
- CODSOFT Internship Guidelines

RESULT

The Tic-Tac-Toe AI project was successfully developed and executed using Python. The game allows players to play interactively through the command-line interface. The program correctly detects valid moves, winning conditions, and Draw situations, providing a smooth and functional gameplay experience.